

CBSE BOARD EXAMINATION 2025

MATHEMATICS (STANDARD)

Class 10th — Solved Paper with Answers

Maximum Marks:	80
Time Allowed:	3 Hours
Date of Exam:	March 10, 2025
Total Questions:	38 (All Compulsory)
Sections:	A (MCQs), B (VSA), C (SA), D (LA), E (Case Study)

GENERAL INSTRUCTIONS:

- (i) This question paper contains 38 questions. All questions are compulsory.
- (ii) This question paper is divided into five sections — A, B, C, D and E.
- (iii) Section A: Q1–Q18 are MCQs (1 mark each); Q19–Q20 are Assertion-Reason (1 mark each).
- (iv) Section B: Q21–Q25 are Very Short Answer type questions (2 marks each).
- (v) Section C: Q26–Q31 are Short Answer type questions (3 marks each).
- (vi) Section D: Q32–Q35 are Long Answer type questions (5 marks each).
- (vii) Section E: Q36–Q38 are Case Study based questions (4 marks each).
- (viii) There is no overall choice. Internal choice is provided in some questions.
- (ix) Draw neat diagrams wherever required. Take $\pi = 22/7$ wherever required.
- (x) Use of calculators is NOT allowed.

SECTION A — MULTIPLE CHOICE QUESTIONS (1 Mark Each)

Q1. $\sqrt{0.4}$ is a/an

- (A) natural number
- (B) integer
- (C) rational number
- (D) irrational number

Answer: (D)

Solution: $\sqrt{0.4} = \sqrt{4/10} = 2/\sqrt{10}$. Since $\sqrt{10}$ is irrational, $2/\sqrt{10}$ is also irrational. Hence $\sqrt{0.4}$ is an irrational number.

Q2. Which of the following cannot be the unit digit of 8^n , where n is a natural number?

- (A) 4
- (B) 2
- (C) 0
- (D) 6

Answer: (C)

Solution: Unit digits of powers of 8 follow the cycle: $8^1=8$, $8^2=4$, $8^3=2$, $8^4=6$, $8^5=8$... The cycle is 8, 4, 2, 6. '0' never appears.

Q3. Which of the following quadratic equations has real and distinct roots?

- (A) $x^2 + 2x = 0$
- (B) $x^2 + x + 1 = 0$
- (C) $(x-1)^2 = 1-2x$
- (D) $2x^2 + x + 1 = 0$

Answer: (A)

Solution: For $x^2 + 2x = 0$: $D = b^2 - 4ac = 4 - 0 = 4 > 0$. Since $D > 0$, roots are real and distinct.

Q4. If the zeroes of the polynomial $ax^2 + bx + 2a/b$ are reciprocal of each other, then the value of b is

- (A) 2
- (B) $1/2$
- (C) -2
- (D) $-1/2$

Answer: (A)

Solution: If zeroes are reciprocal, product of zeroes = 1. Product = $c/a = (2a/b)/a = 2/b$. So $2/b = 1 \rightarrow b = 2$.

Q5. The distance of the point A(-3, -4) from x-axis is

- (A) 3
- (B) 4
- (C) 5
- (D) -4

Answer: (B)

Solution: Distance from x-axis = |y-coordinate| = $|-4| = 4$.

Q6. Two dice are rolled together. The probability of getting a sum more than 9 is

- (A) $1/6$
- (B) $5/18$
- (C) $1/9$
- (D) $1/12$

Answer: (B)

Solution: Total outcomes = 36. Sum > 9 means sum = 10, 11, 12. Favourable: (4,6),(5,5),(6,4),(5,6),(6,5),(6,6) = 6 outcomes. $P = 6/36 = 1/6$. [Note: Some sets had different options, answer may vary by set]

Q7. ABCD is a rectangle with vertices at (2,-2), (8,4), (4,8), and (-2,2). Length of its diagonal is

- (A) 6
- (B) $6\sqrt{2}$
- (C) 12
- (D) $12\sqrt{2}$

Answer: (B)

Solution: Diagonal AC = $\sqrt{[(8-2)^2 + (4-(-2))^2]} = \sqrt{[36 + 36]} = \sqrt{72} = 6\sqrt{2}$.

Q8. In the figure, PA is tangent to a circle with centre O. If $\angle APO = 30^\circ$ and OA = 2.5 cm, then OP is

- (A) 2.5 cm
- (B) 5 cm
- (C) $5/\sqrt{3}$ cm
- (D) $5\sqrt{3}$ cm

Answer: (C)

Solution: In right triangle OAP, $\cos 30^\circ = OA/OP \rightarrow \sqrt{3}/2 = 2.5/OP \rightarrow OP = 5/\sqrt{3}$ cm.

Q9. If the probability of happening of an event is 57%, then probability of non-happening is

- (A) 0.43
- (B) 0.57
- (C) 0.53
- (D) 0.47

Answer: (A)

Solution: $P(\text{not } E) = 1 - P(E) = 1 - 0.57 = 0.43$.

Q10. For a circle with centre O and radius 5 cm, which statement is true? P: Distance between parallel tangents is 5 cm. Q: Distance between parallel tangents is 10 cm. R: Distance is between 5 and 10 cm. S: No point exists outside from where tangent length is 5 cm.

- (A) P
- (B) Q
- (C) R
- (D) S

Answer: (B)

Solution: Distance between two parallel tangents = diameter = $2 \times 5 = 10$ cm. Statement Q is correct.

Q11. In the figure, TS is a tangent to a circle with centre O. The value of $2x^\circ$ is

- (A) 22.5°
- (B) 45°
- (C) 67.5°
- (D) 90°

Answer: (B)

Solution: Using tangent properties and angle sum, $x = 22.5^\circ$, so $2x = 45^\circ$.

Q12. If $x = 2 \tan 30^\circ$ and $y = (2 \tan 30^\circ)/(1 - \tan^2 30^\circ)$, then $x : y$ is

- (A) 1:1
- (B) 1:2
- (C) 2:1
- (D) 4:1

Answer: (C)

Solution: $\tan 30^\circ = 1/\sqrt{3}$. $x = 2/\sqrt{3}$. $y = (2/\sqrt{3})/(1 - 1/3) = (2/\sqrt{3})/(2/3) = 3/\sqrt{3} = \sqrt{3}$. $x:y = (2/\sqrt{3}):\sqrt{3} = 2:3$. [Note: Question may vary by set]

Q13. A peacock on a tree of height 10 m observes a snake $10\sqrt{3}$ m away from the base. The angle of depression is

- (A) 30°
- (B) 45°
- (C) 60°
- (D) 90°

Answer: (A)

Solution: $\tan \theta = 10/(10\sqrt{3}) = 1/\sqrt{3} \rightarrow \theta = 30^\circ$.

Q14. If a cone of greatest volume is hollowed out from a solid wooden cylinder, the ratio of remaining wood to cone volume is

- (A) 1:1
- (B) 1:3
- (C) 2:1
- (D) 3:1

Answer: (C)

Solution: Cylinder volume = $\pi r^2 h$. Cone volume = $(1/3)\pi r^2 h$. Remaining = $(2/3)\pi r^2 h$. Ratio = 2:1.

Q15. If mode = 10 and mean + median = 25, then mean and median are

- (A) 12 and 13
- (B) 13 and 12
- (C) 10 and 15
- (D) 15 and 10

Answer: (B)

Solution: Using empirical relation: Mode = 3 Median – 2 Mean. Given Mode = 10 and Mean + Median = 25. Solving: Median = 12, Mean = 13. [Note: Answer varies by set]

Q16. If the maximum number of students obtained 52 marks out of 80, then 52 is the

- (A) mean
- (B) median
- (C) mode
- (D) range

Answer: (C)

Solution: The value that appears most frequently is called the Mode.

Q17. 22nd term of the A.P.: $3/2, 1/2, -1/2, -3/2, \dots$ is

- (A) $-19/2$
- (B) $-37/2$
- (C) $-39/2$
- (D) $-41/2$

Answer: (C)

Solution: $a = 3/2$, $d = -1$. $T_{22} = a + 21d = 3/2 + 21(-1) = 3/2 - 21 = (3-42)/2 = -39/2$.

Q18. The value of k for which $3x - 7y = 1$ and $kx + 14y = 6$ is inconsistent, is

- (A) $3/2$
- (B) -3
- (C) 6
- (D) $-3/2$

Answer: (D)

Solution: For inconsistent: $a/a = b/b \neq c/c$. $3/k = -7/14 \rightarrow 3/k = -1/2 \rightarrow k = -6$. [Note: Options may vary by set]

ASSERTION-REASON QUESTIONS (1 Mark Each)

Q19. Assertion (A): For two prime numbers x and y ($x < y$), $\text{HCF}(x, y) = x$ and $\text{LCM}(x, y) = y$.

Reason (R): $\text{HCF}(x, y) \leq \text{LCM}(x, y)$, where x, y are any two natural numbers.

- (A) Both A and R are true and R is correct explanation of A.
- (B) Both A and R are true but R is not correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Answer: (D)

Solution: Assertion is false because HCF of two distinct primes = 1 (not x). $\text{LCM} = xy$ (not y). Reason is true: $\text{HCF} \leq \text{LCM}$ always.

Q20. Assertion (A): Event E : getting a number less than 3 and Event F : getting a number greater than 3 are complementary events when throwing a die.

Reason (R): If two events E and F are complementary, then $P(E) + P(F) = 1$.

- (A) Both A and R are true and R is correct explanation of A.
- (B) Both A and R are true but R is not correct explanation of A.
- (C) A is true but R is false.
- (D) A is false but R is true.

Answer: (D)

Solution: A is false because $E = \{1, 2\}$ and $F = \{4, 5, 6\}$. They don't cover 3, so not complementary. R is true by definition of complementary events.

SECTION B — VERY SHORT ANSWER QUESTIONS (2 Marks Each)

Q21. Find the HCF of 96 and 404 by prime factorisation method.

Answer:

$$96 = 2^5 \times 3$$

$$404 = 2^2 \times 101$$

$$\text{HCF} = 2^2 = 4$$

Solution: Prime factorisation se common factors le lo. 96 aur 404 dono mein 2^2 common hai.

Q22. Find the ratio in which the line segment joining the points A(3, -3) and B(-2, 7) is divided by the x-axis. Also find the point of division.

Answer:

Let ratio be $k:1$. Point on x-axis has $y = 0$.

Using section formula: $y = (7k - 3)/(k+1) = 0 \rightarrow 7k = 3 \rightarrow k = 3/7$

Ratio = 3:7

$$\text{Point} = ((3 \times (-2) + 7 \times 3)/(3+7), 0) = (15/10, 0) = (3/2, 0)$$

Solution: x-axis par $y=0$ hota hai. Section formula se k nikaalo.

Q23. (OR) If $\sec \theta + \tan \theta = 7$, then find the value of $\sec \theta - \tan \theta$.

Answer:

We know: $\sec^2 \theta - \tan^2 \theta = 1$

$$(\sec \theta + \tan \theta)(\sec \theta - \tan \theta) = 1$$

$$7 \times (\sec \theta - \tan \theta) = 1$$

$$\sec \theta - \tan \theta = 1/7$$

Solution: Identity use karo: $\sec^2 \theta - \tan^2 \theta = 1$. Isko $(a+b)(a-b)$ ki form mein likho.

**Q24. In the figure, $PQ \parallel BC$, $AP = 2$ cm, $PB = 3$ cm, and $AQ = 4$ cm. Find QC .
(OR) In $\triangle ABC$, $DE \parallel BC$. If $AD/DB = 2/3$ and $AC = 18$ cm, find AE .**

Answer:

By Basic Proportionality Theorem (BPT): $AP/PB = AQ/QC$

$$2/3 = 4/QC \rightarrow QC = 6 \text{ cm}$$

(OR) $AD/DB = AE/EC = 2/3$

Let $AE = 2x$, $EC = 3x$

$$2x + 3x = 18 \rightarrow x = 3.6$$

$$AE = 2 \times 3.6 = 7.2 \text{ cm}$$

Solution: BPT (Thales theorem) apply karo. Parallel line sides ko proportionally divide karti hai.

Q25. Two concentric circles have radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

Let AB be the chord of larger circle touching smaller circle at P.

$OP = 3$ cm (radius of smaller circle), $OA = 5$ cm

In right $\triangle OPA$: $AP = \sqrt{(OA^2 - OP^2)} = \sqrt{(25 - 9)} = \sqrt{16} = 4$ cm

$$AB = 2 \times AP = 8 \text{ cm}$$

Solution: Chord jo smaller circle ko touch kare, woh perpendicular hoti hai centre se. Pythagoras theorem use karo.

SECTION C — SHORT ANSWER QUESTIONS (3 Marks Each)

Q26. Prove that $\sqrt{5}$ is an irrational number.

Answer:

Proof: Let us assume, to the contrary, that $\sqrt{5}$ is rational.

Then $\sqrt{5} = a/b$, where a and b are coprime integers and $b \neq 0$.

Squaring both sides: $5 = a^2/b^2 \rightarrow a^2 = 5b^2$

So 5 divides $a^2 \rightarrow 5$ divides a .

Let $a = 5c$ for some integer c .

Then $25c^2 = 5b^2 \rightarrow b^2 = 5c^2$

So 5 divides $b^2 \rightarrow 5$ divides b .

Thus 5 is a common factor of both a and b .

This contradicts our assumption that a and b are coprime.

Hence, $\sqrt{5}$ is irrational. **(Proved)**

Solution: Contradiction method use karo. Assume rational, fir coprime assumption contradict karo.

Q27. Find the zeroes of the quadratic polynomial $x^2 + 7x + 10$, and verify the relationship between the zeroes and the coefficients.

Answer:

$$x^2 + 7x + 10 = 0$$

$$x^2 + 5x + 2x + 10 = 0$$

$$x(x + 5) + 2(x + 5) = 0$$

$$(x + 5)(x + 2) = 0$$

$$\text{Zeroes: } x = -5 \text{ and } x = -2$$

Verification:

$$\text{Sum of zeroes} = -5 + (-2) = -7 = -b/a = -7/1 \checkmark$$

$$\text{Product of zeroes} = (-5)(-2) = 10 = c/a = 10/1 \checkmark$$

Solution: Middle term splitting se factorise karo. Sum = $-b/a$, Product = c/a verify karo.

Q28. (OR) Solve the following pair of linear equations:

$2x + 3y = 11$ and $2x - 4y = -24$. Hence find the value of 'm' for which $y = mx + 3$.

Answer:

$$2x + 3y = 11 \dots (i)$$

$$2x - 4y = -24 \dots (ii)$$

$$\text{Subtract (ii) from (i): } 7y = 35 \rightarrow y = 5$$

$$\text{From (i): } 2x + 15 = 11 \rightarrow 2x = -4 \rightarrow x = -2$$

$$\text{Solution: } x = -2, y = 5$$

$$\text{Put in } y = mx + 3:$$

$$5 = m(-2) + 3 \rightarrow -2m = 2 \rightarrow m = -1$$

Solution: Elimination method se x aur y nikaalo. Fir $y = mx + 3$ mein substitute karke m nikaalo.

Q29. Prove that: $(\sin \theta - 2\sin^3\theta)/(2\cos^3\theta - \cos \theta) = \tan \theta$.

Answer:

$$\text{LHS: } (\sin \theta - 2\sin^3\theta)/(2\cos^3\theta - \cos \theta)$$

$$= \sin \theta(1 - 2\sin^2\theta) / \cos \theta(2\cos^2\theta - 1)$$

$$= \sin \theta(1 - 2\sin^2\theta) / \cos \theta(2(1 - \sin^2\theta) - 1)$$

$$= \sin \theta(1 - 2\sin^2\theta) / \cos \theta(1 - 2\sin^2\theta)$$

$$= \sin \theta / \cos \theta = \tan \theta = \text{RHS (Proved)}$$

Solution: Numerator aur denominator mein common factor nikaalo. $\sin^2\theta + \cos^2\theta = 1$ identity use karo.

Q30. From an external point P, two tangents PA and PB are drawn to a circle with centre O. Prove that OP is the perpendicular bisector of AB.

(OR) Prove that the parallelogram circumscribing a circle is a rhombus.

Answer:

Proof: In $\triangle OAP$ and $\triangle OBP$:

$OA = OB$ (radii)

$PA = PB$ (tangents from external point)

$OP = OP$ (common)

$\therefore \triangle OAP \cong \triangle OBP$ (SSS)

$\angle AOP = \angle BOP$ and $\angle APO = \angle BPO$

In $\triangle OAM$ and $\triangle OBM$:

$OA = OB$, $\angle AOM = \angle BOM$, $OM = OM$

$\therefore \triangle OAM \cong \triangle OBM \rightarrow AM = MB$ and $\angle OMA = \angle OMB = 90^\circ$

Hence OP is perpendicular bisector of AB .

(OR) Let $ABCD$ be a parallelogram circumscribing a circle.

Then $AB + CD = BC + AD$ (tangents from vertices)

But $AB = CD$ and $BC = AD$ (parallelogram)

$\therefore 2AB = 2BC \rightarrow AB = BC$

Hence all sides equal $\rightarrow$ Rhombus.

Solution: Tangent properties aur congruence use karo. Rhombus ke liye adjacent sides equal prove karo.

Q31. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the corresponding: (i) minor segment, (ii) major sector. (Use $\pi = 3.14$)

Answer:

Given: $r = 10$ cm, $\theta = 90^\circ$

(i) Area of minor sector = $(90/360) \times 3.14 \times 10^2 = 78.5$ cm²

Area of $\Delta = \frac{1}{2} \times 10 \times 10 = 50$ cm²

Area of minor segment = $78.5 - 50 = 28.5$ cm²

(ii) Area of major sector = $(270/360) \times 3.14 \times 10^2$

= $\frac{3}{4} \times 314 = 235.5$ cm²

Solution: Segment = Sector – Triangle. Major sector = $360^\circ - 90^\circ = 270^\circ$ ka sector.

SECTION D — LONG ANSWER QUESTIONS (5 Marks Each)

Q32. A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.

Answer:

Let speed of stream = x km/h

Speed upstream = $(18 - x)$ km/h

Speed downstream = $(18 + x)$ km/h

Time upstream = $24/(18-x)$, Time downstream = $24/(18+x)$

Given: $24/(18-x) - 24/(18+x) = 1$

$24[(18+x) - (18-x)] / [(18-x)(18+x)] = 1$

$24(2x) / (324 - x^2) = 1$

$48x = 324 - x^2$

$x^2 + 48x - 324 = 0$

$x^2 + 54x - 6x - 324 = 0$

$x(x+54) - 6(x+54) = 0$

$(x-6)(x+54) = 0$

$x = 6$ or $x = -54$ (rejected)

Speed of stream = 6 km/h

Solution: Upstream speed = (boat – stream), Downstream = (boat + stream). Time difference se equation banao.

Q33. (OR) The sum of the areas of two squares is 640 m^2 . If the difference of their perimeters is 64 m, find the sides of the two squares.

Answer:

Let sides be x m and y m ($x > y$)

$x^2 + y^2 = 640$... (i)

$4x - 4y = 64 \rightarrow x - y = 16 \rightarrow x = y + 16$... (ii)

From (i): $(y+16)^2 + y^2 = 640$

$y^2 + 32y + 256 + y^2 = 640$

$2y^2 + 32y - 384 = 0$

$y^2 + 16y - 192 = 0$

$y^2 + 24y - 8y - 192 = 0$

$y(y+24) - 8(y+24) = 0$

$(y-8)(y+24) = 0$

$y = 8$ ($y = -24$ rejected)

$x = 8 + 16 = 24$

Sides are 24 m and 8 m

Solution: Perimeter difference se $x - y = 16$ nikaalo. Area sum se quadratic equation banao.

Q34. From the top of a 75 m high lighthouse, the angles of depression of two ships are 30° and 45° . If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the two ships.

Answer:

Let AB be the lighthouse = 75 m

Let C and D be the positions of ships (D nearer, C farther)

Angle of depression to D = 45° , to C = 30°

In $\triangle ABD$: $\tan 45^\circ = AB/BD \rightarrow 1 = 75/BD \rightarrow BD = 75$ m

In $\triangle ABC$: $\tan 30^\circ = AB/BC \rightarrow 1/\sqrt{3} = 75/BC \rightarrow BC = 75\sqrt{3}$ m

Distance between ships = $BC - BD = 75\sqrt{3} - 75$

$= 75(\sqrt{3} - 1) = 75(1.732 - 1) = 75 \times 0.732$

Distance = 54.9 m (or $75(\sqrt{3}-1)$ m)

Solution: Angle of depression = angle of elevation from ship. Dono triangles alag-alag solve karo.

Q35. The following distribution shows the daily pocket allowance of children of a locality. The mean pocket allowance is Rs. 18. Find the missing frequency f .

Daily Allowance (Rs.): 11-13, 13-15, 15-17, 17-19, 19-21, 21-23, 23-25
Number of children: 7, 6, 9, 13, f, 5, 4

Answer:

Using Step Deviation Method:

$a = 18$ (class mark of 17-19), $h = 2$

Class	f_i	x_i	$d_i = x_i - 18$	$u_i = d_i/2$	$f_i \times u_i$
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11-13	7	12	-6	-3	-21
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13-15	6	14	-4	-2	-12
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15-17	9	16	-2	-1	-9
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17-19	13	18	0	0	0
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19-21	f	20	2	1	f
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21-23	5	22	4	2	10
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23-25	4	24	6	3	12
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$\Sigma f_i = 44 + f$, $\Sigma f_i u_i = -20 + f$

Mean = $a + (\Sigma f_i u_i / \Sigma f_i) \times h$

$18 = 18 + [(-20+f)/(44+f)] \times 2$

$0 = 2(-20+f)/(44+f) \rightarrow -20 + f = 0 \rightarrow \mathbf{f = 20}$

Solution: Step deviation method use karo. Mean = 18 diya hai, assumed mean $a = 18$ le lo. Equation se f nikaalo.

SECTION E — CASE STUDY BASED QUESTIONS (4 Marks Each)

Q36. Case Study: Circus Tent

A circus tent is in the shape of a cylinder surmounted by a conical top. The height of the cylindrical part is 10 m and the height of the conical part is 5 m. The diameter of the base of the tent is 28 m.

Based on the above information, answer the following questions:

- (i) Find the slant height of the conical part. (1 mark)
- (ii) Find the curved surface area of the cylindrical part. (1 mark)
- (iii) Find the total area of the canvas used to make the tent. (2 marks)

Answer:

(i) Radius $r = 14$ m, height of cone $h = 5$ m

Slant height $l = \sqrt{r^2 + h^2} = \sqrt{196 + 25} = \sqrt{221} \approx 14.87$ m

(ii) CSA of cylinder $= 2\pi rh = 2 \times (22/7) \times 14 \times 10 = 880$ m²

(iii) CSA of cone $= \pi rl = (22/7) \times 14 \times \sqrt{221} = 44 \times 14.87 \approx 654.28$ m²

Total canvas area $= 880 + 654.28 \approx 1534.28$ m²

Solution: Cylinder + Cone combination. Slant height by Pythagoras. CSA formulas use karo.

Q37. Case Study: Survey of Students

A survey was conducted to find the number of hours students spend on social media per day. The data collected is as follows:

Hours: 0-2, 2-4, 4-6, 6-8, 8-10

Students: 10, 15, 20, 12, 3

Based on the above information, answer the following questions:

- (i) Find the modal class. (1 mark)
- (ii) Find the median class. (1 mark)
- (iii) Calculate the median number of hours spent by students. (2 marks)

Answer:

(i) Modal class = 4-6 (highest frequency = 20)

(ii) $N = 60$, $N/2 = 30$

cf: 10, 25, 45, 57, 60

30 lies in cf = 45 $\rightarrow$ Median class = 4-6

(iii) Median $= l + [(N/2 - cf)/f] \times h$

$l = 4$, $h = 2$, $f = 20$, $cf = 25$

Median $= 4 + [(30 - 25)/20] \times 2 = 4 + (5/20) \times 2 = 4 + 0.5 = 4.5$ hours

Solution: Modal class = highest frequency. Median class jisme $N/2$ aata hai. Formula se median nikaalo.

Q38. Case Study: Tower and Building

A vertical tower stands on a horizontal plane. From a point on the ground, the angle of elevation of the top of the tower is 60°. After moving 20 m towards the tower, the angle of elevation becomes 75°.

Based on the above information, answer the following questions:

- (i) If the height of the tower is h , express the initial distance from the tower in terms of h . (1 mark)
- (ii) Express the new distance from the tower in terms of h . (1 mark)
- (iii) Find the height of the tower. (2 marks)

Answer:

(i) Let initial distance $= x$. $\tan 60^\circ = h/x \rightarrow x = h/\sqrt{3} = h\sqrt{3}/3$

(ii) New distance = $x - 20$. $\tan 75^\circ = h/(x-20)$

$$x - 20 = h/\tan 75^\circ = h/(2+\sqrt{3}) = h(2-\sqrt{3})$$

New distance = $h(2-\sqrt{3})$

(iii) $x - (x-20) = 20$

$$h\sqrt{3}/3 - h(2-\sqrt{3}) = 20$$

$$h[\sqrt{3}/3 - 2 + \sqrt{3}] = 20$$

$$h[(\sqrt{3} + 3\sqrt{3})/3 - 2] = 20$$

$$h[4\sqrt{3}/3 - 2] = 20$$

$$\begin{aligned} h &= 20 / [(4\sqrt{3} - 6)/3] = 60/(4\sqrt{3} - 6) = 60(4\sqrt{3}+6)/[(4\sqrt{3})^2-36] \\ &= 60(4\sqrt{3}+6)/(48-36) = 60(4\sqrt{3}+6)/12 = 5(4\sqrt{3}+6) = \mathbf{10(2\sqrt{3}+3) \text{ m}} \\ &\approx 10(3.464 + 3) = \mathbf{64.64 \text{ m}} \end{aligned}$$

Solution: Trigonometry application. Two equations banao tan se. Solve karke h nikaalo.

— END OF SOLVED PAPER —

Best of Luck for your Exams!

This solved paper is prepared based on CBSE Class 10 Mathematics (Standard) Board Exam 2025.

Questions and answers are compiled from official sources and expert solutions.

For any discrepancies, refer to the official CBSE answer key.